

Week 5

Chain rule (continued)

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

what if there are more than 2 functions

$$\frac{d}{dx} f(g(h(x))) = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x)$$

~~Ex~~
$$\frac{d}{dx} f(2f(3f(x))) = f'(2f(3f(x))) \cdot 2f'(3f(x)) \cdot 3f'(x)$$

wow that's
simple

Ex $g(x) = \left[f\left(\frac{2}{x}\right) \right]^3$

$$g'(x) = 3 \left[f\left(\frac{2}{x}\right) \right]^2 \cdot f'\left(\frac{2}{x}\right) \cdot \left(-\frac{2}{x^2}\right)$$

Trig - Derivatives

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Proof later
can be used in calculations

$$\text{Ex } \lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 7x} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \cdot \frac{7x}{\sin 7x} \right)$$

$$\lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 1, \quad \lim_{x \rightarrow 0} \frac{7x}{\sin 7x} = 1$$

$$\Rightarrow \lim \text{ is } \frac{3}{7}$$

Derivatives

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\cos^2 x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x \quad (\text{exercise})$$

$$\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = -\frac{-\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \cdot \tan x$$

Note: the trig function d.h formulas assume the argument is in radians

if x is in degrees what is the angle in radians

$$x \text{ in degrees} \longrightarrow \frac{\pi x}{180} \text{ radians}$$

if x is in degrees,

$$y = \sin\left(\frac{\pi x}{180}\right) \Rightarrow y' = \frac{\pi}{180} \cdot \cos\frac{\pi x}{180}$$

$$y' \Big|_{x=45} = \frac{\pi}{180} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360}$$

$$y_0 = y(45^\circ) = \sin(45) = \frac{\sqrt{2}}{2}$$

Equation of tangent line:

$$y - \frac{\sqrt{2}}{2} = \frac{\sqrt{2}\pi}{360} (x - 45)$$

$$\lim_{x \rightarrow 0} x^2 \csc x \cot x$$

$$= \lim_{x \rightarrow 0} x^2 \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^2 \cos x$$

↓ ↓
1 1

$\Rightarrow$ lim is 1.